

Shuai Ma

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Education

Seattle University

Seattle, WA, United States

Albers School of Business & Economics

M.S. in Business Analytics, STEM-designated

2018–2020

- Relevant coursework: programming for business analytics, applied econometrics, statistical learning, forecasting, big-data analytics, data management, data visualization, and data architecture.
- Capstone: *Influential Factors on Residential Property Prices Using Data-Mining Methods*; analyzed Pierce County housing transactions using statistical and machine-learning methods.

University of Cincinnati

Cincinnati, OH, United States

College of Design, Architecture, Art, and Planning

M.S. in Architecture, research-based program; research and thesis focused on urban development

2009–2011

- Research training: urban theory, research methods, pre-thesis research, social-science research for planners, land-use control, international development policy, and thesis-based scholarly writing.
- Thesis: *Impact of High-Speed Rail Station Development on Surrounding Urban Form: The Beijing–Tianjin Case*.

Hebei University of Technology

Tianjin, China

Project 211 and Double First-Class university

B.Eng. in Urban Planning and Design

2003–2008

- Undergraduate planning foundation: planning studios, regional planning, urban design, urban economics, urban geography, transportation planning, urban ecology, planning code, and spatial information technology.
- NACES course-by-course evaluation: equivalent to a U.S. bachelor's degree in Urban and Regional Planning.

Research Interests

Housing and urban policy; place-based public investment; housing and land-market capitalization; causal inference and quasi-experimental urban policy evaluation; spatial spillovers, affordability, neighborhood sorting, and distributional effects; climate adaptation and flood mitigation; GIS, spatial econometrics, and urban data science.

Research Papers and Writing Samples

Hazard or Amenity? Flood Risk, Waterway Proximity, and Neighborhood Inequality in Housing Prices

2025–Present

Sole-authored PhD writing sample; empirical study of Pierce County, Washington

- Examines how flood risk and waterway proximity are jointly capitalized into housing prices, and how this capitalization varies across neighborhoods differing in income, school quality, and crime, using 47,763 single-family sales linked to parcel records, FEMA flood-risk layers, and neighborhood indicators.
- Estimates hedonic regression models with tract-clustered standard errors and conducts spatial analysis to distinguish flood-risk effects from waterfront-amenity value.
- Provides the empirical foundation for a broader research agenda on how place-based public investments reshape housing markets, generate spatial spillovers, and distribute benefits and costs across communities.

Strategic Management's Foundational Silence

2025–Present

First-author manuscript in revision, coauthored with Rubina Mahsud and Greg Prussia; developed from a 2020 Strategic Management Society conference proposal

- Built and cleaned a corpus of 3,202 Strategic Management Journal articles and applied classical natural language processing methods, local open-source large language models, and cloud-based models to analyze the field's dominant vocabulary.
- Develops the concept of “foundational silence” to argue that poverty, war, disease, hunger, homelessness, and related human concerns should be treated as central social contexts rather than peripheral externalities.

Conference Presentations

Strategic Management Society 2060: From Effectiveness to Impact, from Success to Significance

October 2020

Strategic Management Society Annual Conference, Virtual

Mahsud, R., and Ma, S. Presented by Shuai Ma.

Selected Research and Analytical Projects

- Pro Forma Urbanism: Financial Feasibility Modeling and Land Value Capture in Shenzhen Urban Renewal** 2025
Sole-authored research brief; reconstructs a developer feasibility model as the interface where planning rules become built form
- Reconstructed the dual-track (static and dynamic) project appraisal as reproducible spreadsheets, reproducing the source's reported IRR of 27.45% and NPV of approximately ¥2.44 billion.
 - Reverse-engineered the FAR-graduated, tenure-contingent land-pricing algorithm and showed it operates as a land value capture instrument: roughly 86% of the ¥4.66 billion premium falls on net-new density, raising the case equity floor land price about 50% above benchmark.
- Empirical Analysis of Guangzhou Residential Rental Markets** 2021
Sole-authored hedonic study; shows dwelling size and central location are jointly capitalized into asking rents
- Collected and cleaned approximately 3,000 public rental listings and estimated a log-log hedonic rent model, addressing right-skew through log transformation and reporting HC1 heteroskedasticity-robust standard errors after a Breusch-Pagan test.
 - Found floor area the dominant structural correlate (rent elasticity 0.715) and a center-oriented spatial gradient: conditional on dwelling attributes, listings rent roughly 51% higher in Tianhe than in reference-district Baiyun.
- Operationalizing Highest and Best Use: A GDV-Oriented Integer-Programming Approach to Product-Mix Optimization for Business-Park Development** 2021
Initiated and led an internal real-estate analytics study at Ping An Group, combining field market research, regulatory analysis, and Python integer programming
- Surveyed 11 competing business parks in Foshan's Nanhai District to parameterize 16 product typologies, then formulated and solved an integer linear program in Python (PuLP / COIN-OR CBC) maximizing gross development value (GDV) under floor-area-ratio and plot-coverage constraints, returning a certified-optimal building mix in seconds.
 - Demonstrated that the value-maximizing mix pairs multi-floor manufacturing with offices and excludes low-margin warehouses, then ran sensitivity scenarios quantifying the GDV cost of adding warehouses to improve market absorption, translating Highest-and-Best-Use logic into a reproducible, data-driven development strategy.
- Engineering Controlled Flexibility: New-type Industrial Land (M0) and the Governance of Innovation Space in the Pearl River Delta** 2021
Sole-authored comparative policy analysis; reads M0 as controlled flexibility via conditional property rights
- Coded M0 regulatory provisions across seven Pearl River Delta cities into a six-dimension matrix (admission, intensity, functional mix, transfer and tenure, value capture, monitoring), triangulated against a 2021 market scan.
 - Found Shenzhen pairs the lowest FAR floor with the highest saleable-area cap (65% vs. 49–50%), offsetting tradability through the Industrial Block-Line; for-sale supply drifted toward saleable product, with over 1,800 R&D units listed against factory stock in only two projects.
- Land Use as a Hydrological Variable: A Site-Conditioned Framework for Low-Impact Development** 2011–2017
Sole-authored retrospective multi-case study; argues land use is a first-order determinant of stormwater design
- Read six sponge-city and water-environment commissions across Shanghai, Shenzhen, Kunshan, and Jinjiang as a comparative case series along a watershed-to-plot scale ladder, pairing SWMM simulation, rational-method and water-balance calculation, and multi-criteria storage trade-offs.
 - Derived five land-use-conditioned design propositions linking terrain to source control, low permeability to detention, density to storage economics, industrial land to clean-dirty separation, and basin scale to hydromorphological restoration.
- Jiangsu Provincial Low-Carbon Urban System Strategy** 2011
Core project-team member, Arup; province-scale carbon-accounting and policy-scenario analysis
- Helped build a multi-sector accounting model integrating activity data across land use, buildings, industry, transportation, water, waste, and ecological space for a 13-city provincial urban system.
 - Produced baseline and policy-scenario estimates for 2005–2030 using per-capita and per-GDP intensity indicators, translating spatial and administrative data into planning-relevant quantitative evidence.
- ## Technical Skills
- Econometric and Quantitative Methods**
Applied econometrics; hedonic price modeling; OLS with clustered standard errors; model selection and multicollinearity diagnostics; statistical learning and forecasting; scenario and baseline modeling; linear programming and optimization
- Spatial and Data Methods**
GIS, spatial analysis, spatial data integration, geospatial data processing, data visualization, web-scraped and administrative data
- Programming and Software**

Professional Experience

Founder & Principal Researcher, Beamlinked Consulting Co., Ltd.

2023–Present

- Founded an independent research and advisory practice focused on housing and real estate markets, land-use policy, and urban development, with project engagements in Shandong, Guangzhou, Chongqing, and other regions across China.
- Conducted market, policy, and spatial analyses for housing, business-park, and urban-development projects, integrating administrative information, online market data, land-use controls, and GIS to assess market demand, development feasibility, and product positioning.
- Developed quantitative and spatial workflows using Python, R, GIS, and administrative and web-collected data to support research and advisory assignments.

Vice Director, Property Asset Management, Ping An Group

2020–2023

Planning, research, and investment analysis lead within a real estate business unit

- Led housing and real estate market research and investment analysis for a business unit, producing recurring analytical reports for senior management on housing markets, urban policy, infrastructure investment, and industrial-park planning.
- Initiated quantitative studies on product positioning and highest-and-best-use optimization, evaluating how regulatory and market constraints affected product mix and gross development value.
- Conducted feasibility, market, and investment analysis for five industrial-park development projects with a combined planned investment of approximately RMB 3.75 billion, two of which are now operational.

Director, Planning and Development, China Fortune Land Development

2017–2018

- Directed planning and design across two public-private-partnership new-town districts (~3,200 acres each), spanning land use, urban design, road and infrastructure networks, flood-control and water systems, residential neighborhoods, parks and public facilities, within development contexts involving land acquisition and resident resettlement.
- Coordinated large-scale development across public and private actors, integrating land-use regulation, infrastructure provision, and real estate development.
- Co-developed a group-wide development guideline adopted as a standard for new real estate projects.

Project Manager, AtkinsRéalis / Atkins

2013–2017

- Led planning and design teams across town and master planning, business and industrial parks, commercial complexes, and urban development projects.
- Directed applied research on environmental and performance-based analysis (microclimate, energy, and building-physics simulation), applying quantitative modeling to development and design decisions.
- Advised project teams on green-building and sustainability standards (LEED, BREEAM, WELL, and equivalents).

Urban Planner, Arup

2011–2013

- Contributed to urban regeneration and redevelopment projects, including the Shanghai Expo Urban Best Practices Area (UBPA) and Knowledge & Innovation Community (KIC) renewal, alongside waterfront, transit-oriented, and regional planning with spatial and GIS-based analysis.
- Contributed to the Jiangsu Provincial Low-Carbon Urban System Strategy, supporting carbon-audit modeling and province-scale scenario analysis.
- Supported a National Basic Research Program study and contributed research materials incorporated into two published department-led books: *Low-Carbon Ecological Space: Rethinking Cross-Dimensional Planning Methods* and *Study on the Economics of Green Buildings in China*.

Assistant Planner, Part-Time, Tongji University Urban Planning and Design Institute

2007–2009

- Supported regional planning, comprehensive urban development planning, tourism planning, historic-district design, and industrial-park planning projects.

Certifications and Professional Credentials

LEED AP Neighborhood Development, U.S. Green Building Council

Honors and Awards

- Dean's Honor List, Albers School of Business and Economics, Seattle University, Fall 2018.
- Best Graduate Design Award, Hebei University of Technology, 2008.
- Second Prize, National Undergraduate Urban Design Competition, Urban Planning Discipline Steering Committee, Ministry of Construction, China, 2007.